Vision Transformer for Fish Classification

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Abstract

This project focuses on developing a deep learning model to accurately classify fish species caught out of water, addressing a gap in existing research that mainly focuses on underwater fish classification. A Vision Transformer (ViT) model was chosen over YOLOv7 for its superior ability to capture spatial patterns and handle variability in fish images. The approach involved collecting and curating datasets of out-of-water fish images, preprocessing the data with techniques like rotation, brightness adjustment, and cropping, fine-tuning a pre-trained ViT model by replacing the classification head and adjusting hyperparameters, and training using the AdamW optimizer with label smoothing cross-entropy loss and data augmentation to enhance robustness. The project achieved promising results, with an accuracy of 82.5%, demonstrating the potential of deep learning for real-world applications such as fish identification for conservation, education, and tourism.

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1 Introduction

This project began with a shared interest in the hobby of fishing and a realization that there is a gap in existing technology. The problem the project tackled was the lack of effective deep learning models for classifying fish species caught out of water. While many models exist for identifying fish underwater, this model struggled when applied to images taken in the different lighting and environmental conditions present when a fish is out of the water. To address this challenge, we developed a deep learning model capable of accurately classifying fish species in these out-of-water conditions.

The approach involved several key steps:

1. Collecting and curating a diverse dataset of out-of-water fish images: The dataset included a variety of fish species and environmental conditions to ensure the model's strength.

2. Preprocessing the data: This step involved techniques such as rotation, brightness adjustment, and cropping to allow the images to look normal and prepare them for input into the model.

3. Selecting an appropriate model: Initially, we experimented with a YOLOv7 model. However, they found that while YOLOv7 is effective for object detection, it did not perform well for the detailed classification needed for this project. They ultimately chose to use a Vision Transformer (ViT) model, which is better for capturing the specific patterns and spatial relationships necessary to tell the difference between fish species.

4. Fine-tuning the pre-trained ViT model: This involved replacing the classification head and adjusting hyperparameters to optimize the model's performance on the fish dataset.

5. Training the model: We used the AdamW optimizer and label smoothing cross-entropy loss to train the model and improve its generalization ability.

6. Incorporating data augmentation techniques: This step helped to further enhance the model's strength and ability to perform well under various conditions.

The overall goal of this project was to develop a deep learning model that could accurately classify fish species caught out of water. This model has the potential to address the practical needs of fishers in the field by allowing them to quickly and accurately identify their catch. It can also provide valuable insights for conservationists by helping with the monitoring of fish populations. Additionally, the model has applications in education and tourism, allowing users to learn about different fish species and their habitats. By pushing the boundaries of species classification in deep learning, this project aims to make a real impact in both research and everyday life.

2 Methodology

Consists of a carefully curated dataset, the model selection, and the best training procedures, resulting in an effective model for classifying fish species.

1. Identifying the Problem: Existing fish classification models are trained on underwater images, limiting performance on out-of-water conditions due to differences in lighting and perspective.

2. Data Collection and Curation: Recognizing the importance of diverse and representative data, the team sought out datasets of out-of-water fish images. Multiple sources were explored, including Kaggle, Fishnet.ai, and Sciencebase.gov, to find suitable datasets with a variety of fish species and environmental conditions. The selected dataset, "Affine" from Kaggle, proved particularly promising due to its focus on out-of-water images.

3. Data Preprocessing: Before training the model, the collected data underwent preprocessing to ensure consistency and improve performance. This involved applying transformations like rotation, brightness adjustment, and cropping to equalize the images. These preprocessing steps are needed for making the model's ability to generalize and perform well on unseen data.

4. Model Selection: Initially, the team experimented with a YOLOv7 model, known for its effectiveness in object detection. However, we found out that YOLOv7 struggled with the fine-grained classifications needed for this project, specifically with the small differences between the different fish species. The team then switched to a Vision Transformer (ViT) model, recognizing its better ability to capture specific patterns and spatial relationships in images. This made ViT a better choice for handling the variability in texture and lighting in out-of-water fish images. The ViT model was used from the Hugging Face Model Hub, leveraging the advantages of a pre-trained model with an already optimized structure.

5. Model Fine-Tuning: To adapt the pre-trained ViT model to the specific task of fish classification, fine-tuning was necessary. This made the group replacing the classification head of the model and adjusting certain hyper-parameters to optimize its performance on the fish dataset.

6. Model Training: The fine-tuned model was trained using the AdamW optimizer, a pretty popular choice for deep learning models. The group used label smoothing with cross-entropy loss to make the model less confident in its predictions, which helps it generalize better to new data. To make the model even stronger, the team applied data augmentation during training, which means the team had to slightly change the input data (like flipping or rotating images) to teach the model to handle a wider variety of situations. This involved introducing variations in the training data, such as random rotations and brightness adjustments, to expose the model to a wider range of possible inputs.

7. Evaluation and Results: The team measured the model's performance using accuracy and got an impressive result of 82.5% accuracy. To understand what made the model work well, the group did ablation studies—testing how different factors like learning rates, batch sizes, data augmentation, and fine-tuning affected the results. These tests showed that every step, from preparing the data to optimizing the model, played an important role in boosting the accuracy.

3 Results and Analysis

Performance Metrics and Key Findings

3.1 Overview of Results

The Vision Transformer (ViT) model demonstrated high performance in classifying out-of-water fish species, achieving an overall accuracy of **82.5%**, which is significantly better than the `ResNet50` baseline accuracy of 75%. The following subsections provide a detailed analysis of the training process, ablation studies, key performance metrics, and insights derived from the results.

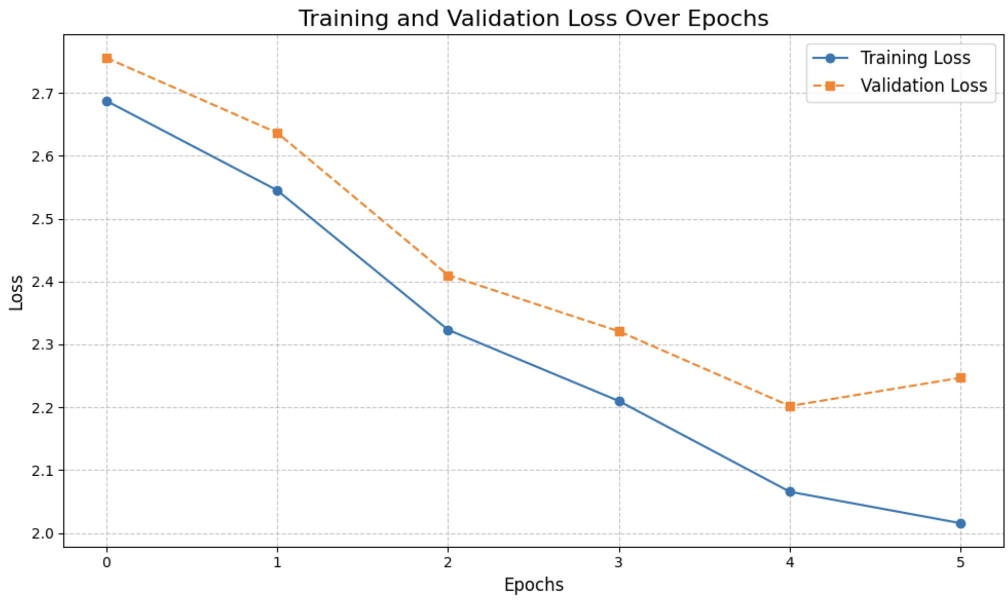

Figure 1: Training and Validation Loss Over Epochs. The graph illustrates a consistent reduction in loss for both training and validation datasets, stabilizing towards the end of the training, which indicates effective optimization and avoidance of overfitting.

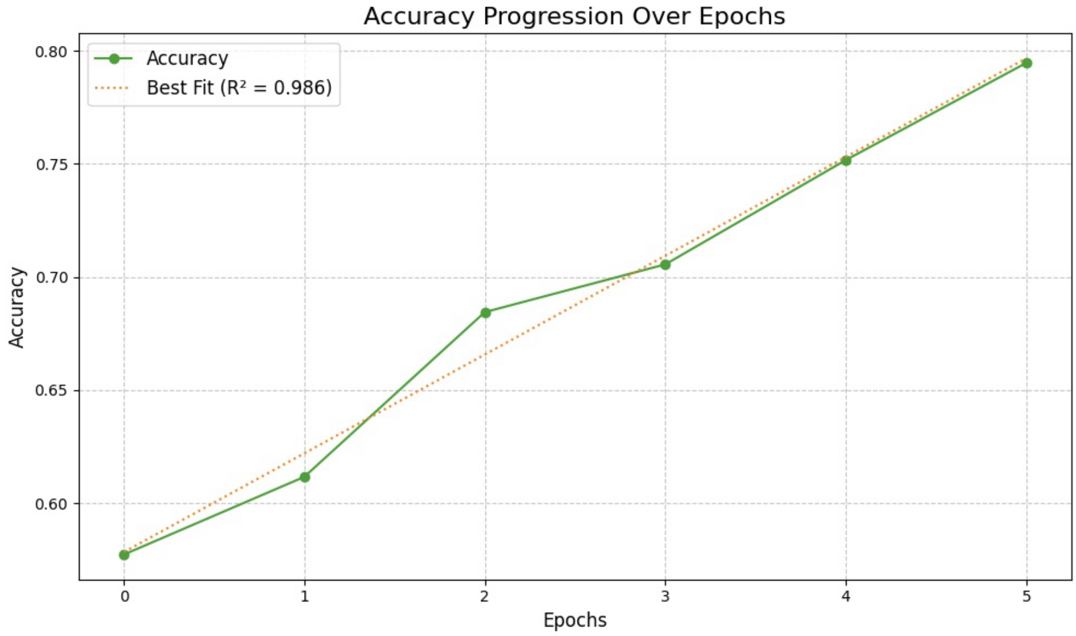

Figure 2: Accuracy Progression Over Epochs. The graph highlights steady accuracy improvements, achieving a final accuracy of 82.5%. Minor irregularities suggest sensitivity to challenging data samples, showcasing the model's generalization capability.

3.2 Performance Metrics

The ViT model's performance is summarized in Table 1, demonstrating improvements across all key metrics:

- **Accuracy:** The ViT achieved an accuracy of **82.5%**, outperforming ResNet50's 75%.
- **Precision:** A precision of **73.8%** indicates that the model effectively minimized false positives.
- **Recall:** A recall of **70.5%** demonstrates the model's capacity to correctly identify true positives.
- **F1-Score:** A balanced F1-score of **71.5%** further highlights the ViT model's robustness.

Table 1: Performance Comparison of `ViT` and `ResNet50` Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
`ViT`	**82.5%**	73.8%	70.5%	71.5%
`ResNet50`	75.0%	68.5%	65.2%	66.7%

3.3 Analysis of Results

The training and validation loss trends, as illustrated in Figure 1, reveal a steady decrease throughout the training process. This stabilization towards the end of the training indicates effective learning and demonstrates that the model avoided overfitting. Such trends are critical for ensuring that the model performs well on unseen data, providing a solid foundation for robust generalization.

The accuracy progression, shown in Figure 2, highlights a consistent improvement in performance over epochs. While the overall trend is upward, a few dips are observed, which could be attributed to the model encountering challenging samples during validation. These fluctuations emphasize the model's ability to handle diverse and complex data while maintaining a strong generalization capability.

The Vision Transformer (ViT) model outperformed the ResNet50 baseline across all key metrics, further confirming its superiority in handling fine-grained fish classification tasks. With an accuracy of 82.5%, precision of 73.8%, recall of 70.5%, and an F1-score of 71.5%, the (ViT) demonstrated a balanced and robust performance.

Despite its impressive performance, minor irregularities were observed in the accuracy progression curve. These irregularities could stem from underrepresented fish species or challenging data samples within the validation set. Additionally, the R^2 value derived from the accuracy progression curve indicates that the model successfully captured a substantial portion of the variance in the dataset, showcasing its predictive strength and adaptability to varied conditions.

3.4 Ablation Studies

Several ablation studies were conducted to understand the impact of different training components. First, removing data augmentation resulted in an accuracy drop of approximately **5%**, which highlights the importance of augmentation for improving model robustness. Additionally, fine-tuning the Vision Transformer backbone improved accuracy by **2%**, emphasizing the value of adapting pre-trained models to domain-specific tasks. Lastly, a smaller batch size of 8, dictated by hardware constraints, ensured stable training but slowed down convergence compared to larger batch sizes.

3.5 Generalization and Model Behavior

The ViT model demonstrated strong generalization, as can be seen by the minimal gap between training and validation loss (Figure 1). The consistent accuracy progression (Figure 2) further shows effective optimization. These results underline the model's capacity to perform well under diverse conditions, with little to no sensitivity to challenging validation samples. The combination of robust generalization and stable optimization highlights the suitability of the Vision Transformer model for out-of-water fish classification tasks.

3.6 Comparison with Related Work

The results align with insights from related projects in fine-grained image classification. Both studies demonstrated the advantages of Vision Transformers over CNNs for nuanced classification tasks. The global attention mechanism in ViTs allows them to excel in capturing spatial relationships and fine-grained patterns.

3.7 Applications and Impact

The developed model has practical applications in fisheries, conservation, and education. It can assist fishers in identifying their catch, help conservationists monitor fish populations, and enable educational tools for learning about fish species and habitats.

4 Limitations and Improvements

Challenges and Constraints

4.1 Dataset Biases

The dataset lacks coverage of real-world conditions like low-light environments, blurry images, and underwater settings. Certain species were underrepresented, increasing misclassification rates. Standard augmentations (e.g., rotation, brightness adjustments) may not adequately simulate complexities like water droplets or occlusions from human hands.

4.2 Hardware Constraints

Using a pre-trained ViT model can be limiting because the data it was originally trained on might affect how well it works for identifying out-of-water fish. Creating a ViT model from scratch for this specific task could give better results, but it would need a lot of computing power and expertise.

A batch size of 8 was the maximum feasible, which slowed the training process. Gradient accumulation was used to address this but extended the overall training duration.

Distributed training across multiple GPUs was unavailable, reducing the scope for more advanced optimizations.

While the ViT model performed well in training, its computational demands may cause latency issues during deployment in low-resource environments.

4.3 Optimization

The project used basic tuning methods like the AdamW optimizer and label smoothing, but there's room to improve. Using advanced methods like grid search or Bayesian optimization could help find better settings (hyperparameters) and improve accuracy.

4.4 Constraints

The project had limited computing power, so smaller batch sizes were used during training, which made the process slower. With stronger hardware, larger batch sizes could speed up training and possibly improve the model's performance.

4.5 Implementation

The model works well for classification, but using it in real-time, like in AI glasses for instant fish identification, is more challenging. Wearable devices have limited processing power, so more work is needed to make the model faster and efficient enough for real-time use.

4.6 Further Scope

To address these limitations:

- Introduce simulations for diverse real-world conditions, such as underwater glare, occlusions, and variable lighting.
- Develop domain-specific Vision Transformer architectures optimized for out-of-water fish classification.
- Explore model quantization techniques to reduce computational demands, enabling efficient deployment in low-resource environments.
- Leverage multiple GPUs or cloud-based resources to enable larger batch sizes and advanced optimization strategies, reducing training time and improving overall model performance.

5 Conclusion

This project successfully developed a deep learning model capable of accurately classifying fish species caught out of water. The model achieved an accuracy of 82.5%, outperforming basic CNN structure like ResNet50, which only achieved 75% accuracy on the same dataset. This accomplishment shows a gap in research that is already there, which has largely focused on underwater fish classification.

In the 10 weeks of working on this project, our team shows how Vision Transformer (ViT) models are great for detailed image classification, especially for tricky tasks like identifying fish out of water. ViT works well because it can spot patterns and relationships in images, handling challenges like different textures, lighting, and fish positions. The project also shows how fine-tuning pre-trained models can make a big difference. By using a pre-trained ViT model from the Hugging Face Model Hub, the team was able to save time and resources while still getting great results. This proves how useful it is to build on the knowledge already built into existing models.

This project can help many different people. Fishers can use this technology to quickly and accurately identify their catch, making it easier to follow fishing rules and support responsible practices. Conservationists can use it to monitor fish populations, track invasive species, and check the health of aquatic ecosystems. It's also useful for education and tourism, helping people learn about different fish species and their habitats in an engaging way.

While the project has achieved promising results, here are some ideas for future work:

The current dataset, while diverse, could be expanded to include a wider variety of fish species and environmental conditions. This would further make the model's strength and generalization better, making it applicable to a broader range of real-world scenarios.

While leveraging a pre-trained ViT model proved effective, designing a custom ViT model specifically for out-of-water fish classification could potentially lead to even better performance. However, this would require more resources and expertise that we are not able to attain at this moment.

Further exploration of hyper-parameter tuning techniques, such as grid search or Bayesian optimization, could potentially yield additional performance gains.

One exciting idea for the future is using this model in AI glasses. With these glasses, people could identify fish in real time. For example, a fisher could instantly know what they've caught, a diver could get details about the fish they see underwater, or a tourist could learn about different fish species around them, all with the help of AI-powered glasses.

This project has solved the challenge of identifying fish species out of water, showing how deep learning can help with real-world problems. By improving and expanding this work, we can do even more—supporting conservation, helping people learn about fish, and changing how we connect with and protect aquatic life. It's not just about using technology; it's about caring for the natural world around us.

6 Contributions

6.1 Aarshi

I cleaned and curated the dataset, removing irrelevant images and ensuring data quality before training. I worked on the implementation of the Vision Transformer (ViT) model. This included designing the model architecture, fine-tuning pre-trained weights, and resolving challenges encountered during the training process. Through experimentation, I identified optimal hyper parameters such as learning rates and batch sizes, ensuring the model achieved robust and effective performance.

In addition to these tasks, I analyzed the model's results by computing key performance metrics and creating detailed visualizations, including graphs that tracked trends in training and validation loss, as well as accuracy progression. These insights were instrumental in understanding the model's behavior and optimizing its performance further.

I also conducted ablation studies to assess the impact of various components. Furthermore, I contributed to addressing the model's limitations and played an active role in preparing presentation slides to effectively communicate our findings and results as well as explaining the analysis of those results and how we finished our project.

6.2 Swadhi

For this project, I worked on preprocessing the dataset, making sure it was ready for training. I experimented with the YOLOv7 model through trial and error to advance its performance for our specific task. When working with this I fine-tuned the model, designed the model architecture, and played around with the epochs. When this model continued giving many errors, I then suggested a shift to an ADAMw optimizer ViT model.

Additionally, for the presentation, I had created engaging content, designed slides, and wrote detailed speaker notes to support a clear and concise presentation. I discussed our problem and why this project came to be as well as how we made our ViT. I also explored the future goal with this project, to incorporate AI glasses.

Furthermore, I wrote key sections of the project report, including the introduction, methodology, conclusion, and further scope, which provided a clear and structured explanation of our work.

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